

Integration by Substitution



u-substitution

1 Evaluate each integral using substitution:

a $\int 2x(x^2 + 1)^5 dx$

b $\int \frac{x}{\sqrt{x^2 + 9}} dx$

c $\int \cos(3x) dx$

2 Evaluate $\int x^2 e^{x^3} dx$.

Definite integrals with substitution

3 Evaluate $\int_0^2 x\sqrt{x^2 + 1} dx$ exactly, changing the limits with $u = x^2 + 1$.

4 Evaluate $\int_0^{\pi/2} \sin^3 x \cos x dx$.

5 Evaluate $\int_1^e \frac{\ln x}{x} dx$.